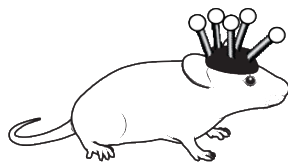


Animal behavior analysis in python

Day 5 Neural correlation and free exploration

Miao Wang wang.miao@campus.lmu.de

Dataset

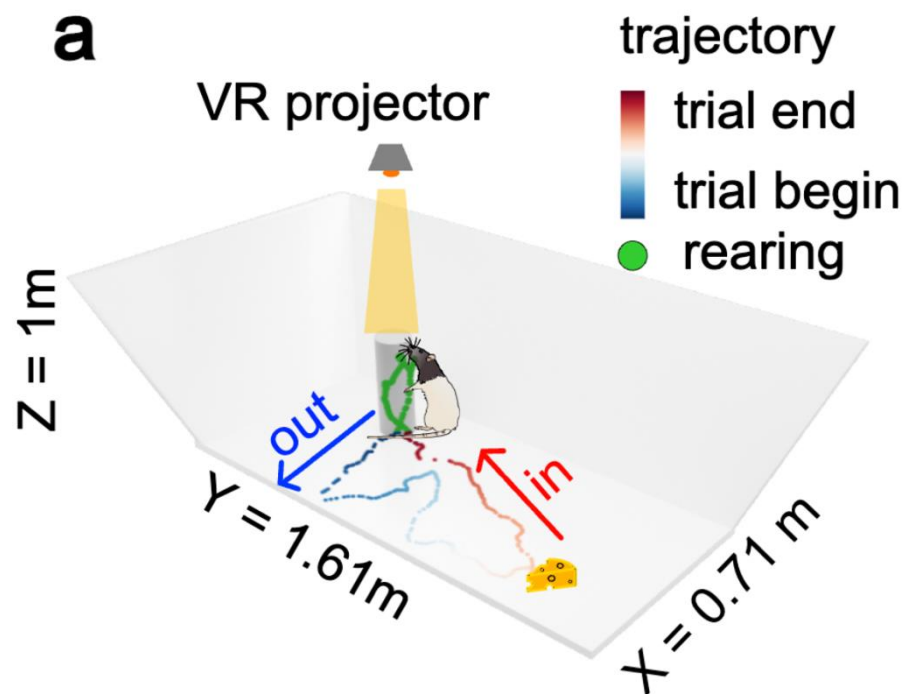


Rat navigation:

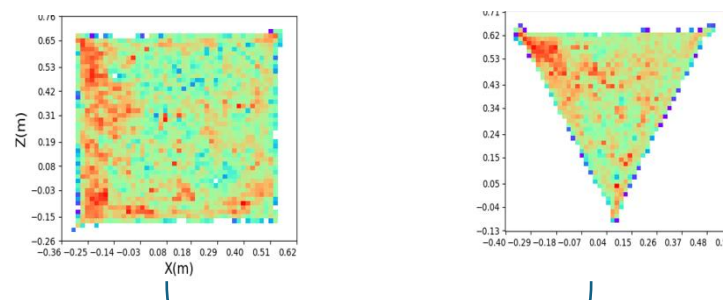
Trial based, 3D trajectory tracking

Mice open field foraging
3D trajectory tracking+ video + head
direction cells

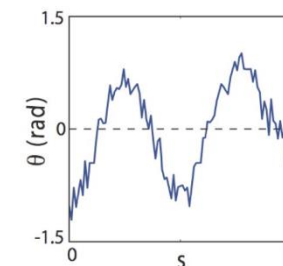
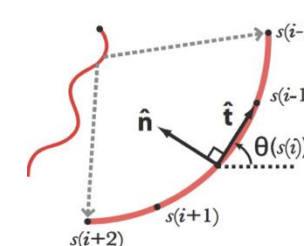
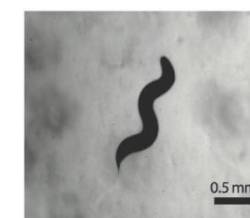
C. Elegance data
Angle of 100 segments



(Fabian Stocek)



(Adrian J Duszkievicz)



A Markovian dynamics for *Caenorhabditis elegans* behavior across scales

Antonio C. Costa^{a,1,2}, Tosif Ahamed^b, David Jordan^c, and Greg J. Stephens^{a,d}

Direction1: connection to HD cells

Day5_Neural_Behavioral_Integration.ipynb

NWB: neuroscience without border



Pynapple

<https://pynapple.org/examples.html>

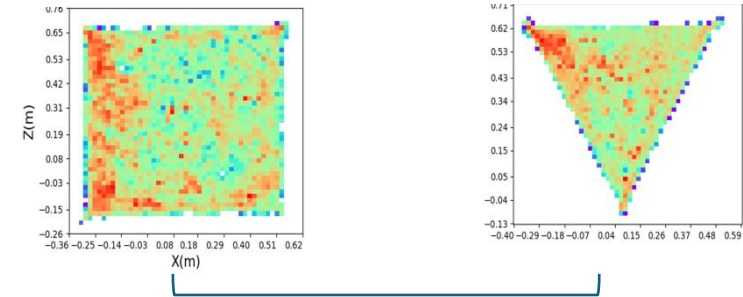
Mice open filed foraging
3D trajectory tracking+ video + head
direction cells

```
print(this_session)
```

✓ 0.0s

area-pos_sub-normalA3717_ses-201021_epoch-0

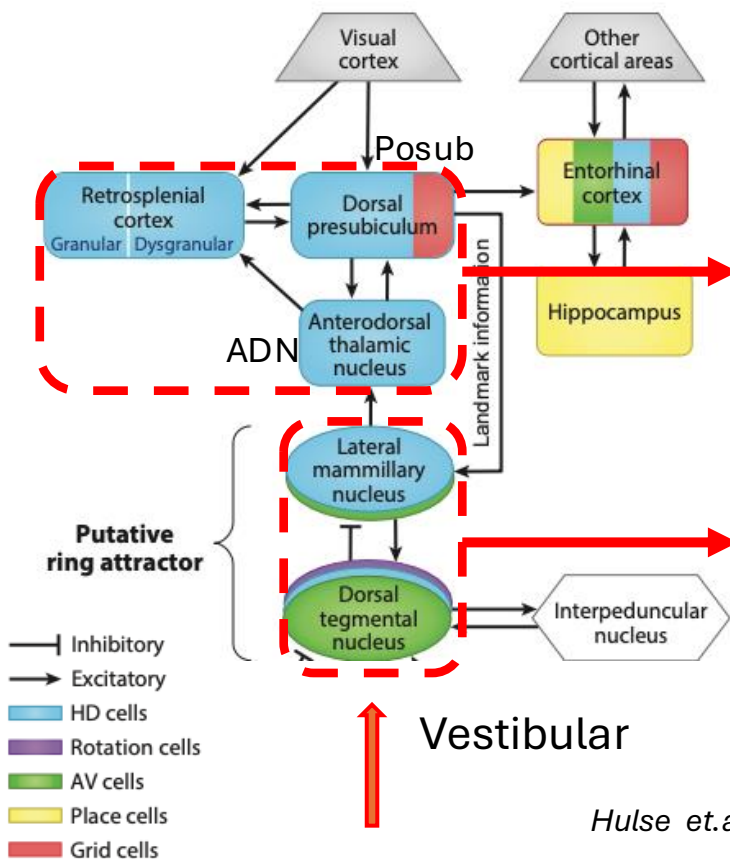
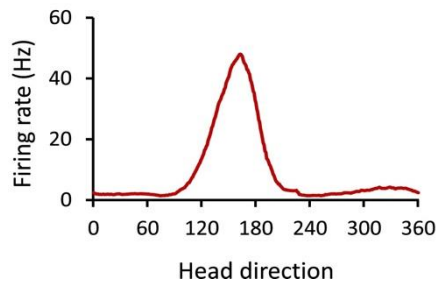
Keys	Type
units	TsGroup
epoch	IntervalSet
z	Tsd
y	Tsd
x	Tsd
rz	Tsd
ry	Tsd
rx	Tsd
distance_border	Tsd



Recording from Posub

Modulation on head-direction(HD) representation

HD cells: stable tuning with peak(s) to certain head direction(s). *Dudchenko et al. (2019).*



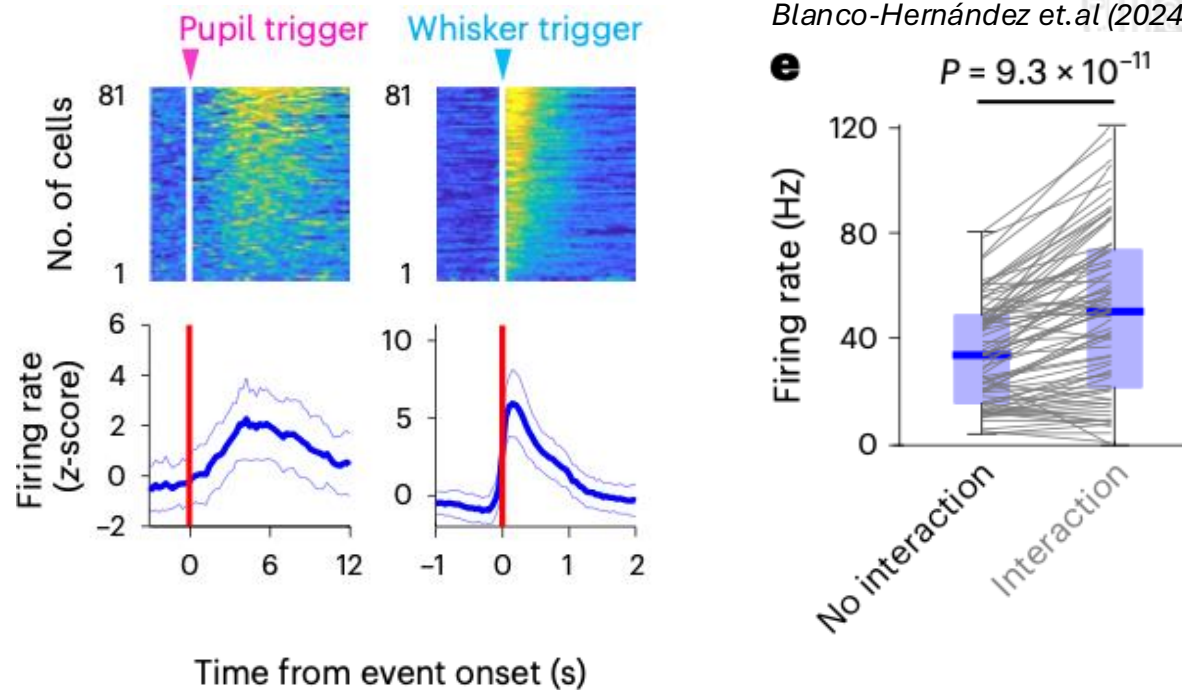
Visual input, context-dependent modulation

Conjunctive representation: angular head velocity, head pitch

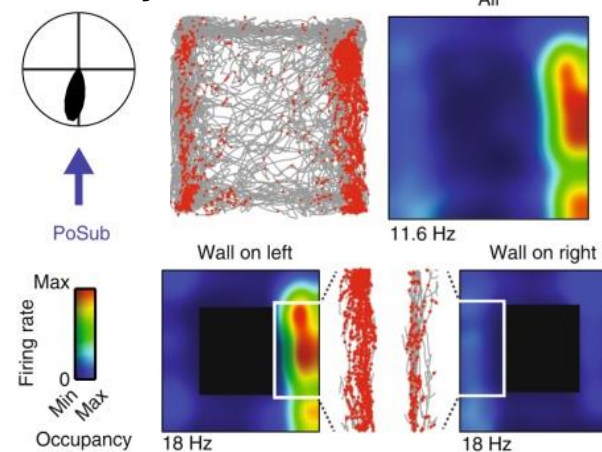
Hulse et al (2020).

Modulated by sensation and social context in ADN

Blanco-Hernández et.al (2024).



Modulated by spatial context in ADN and Posub



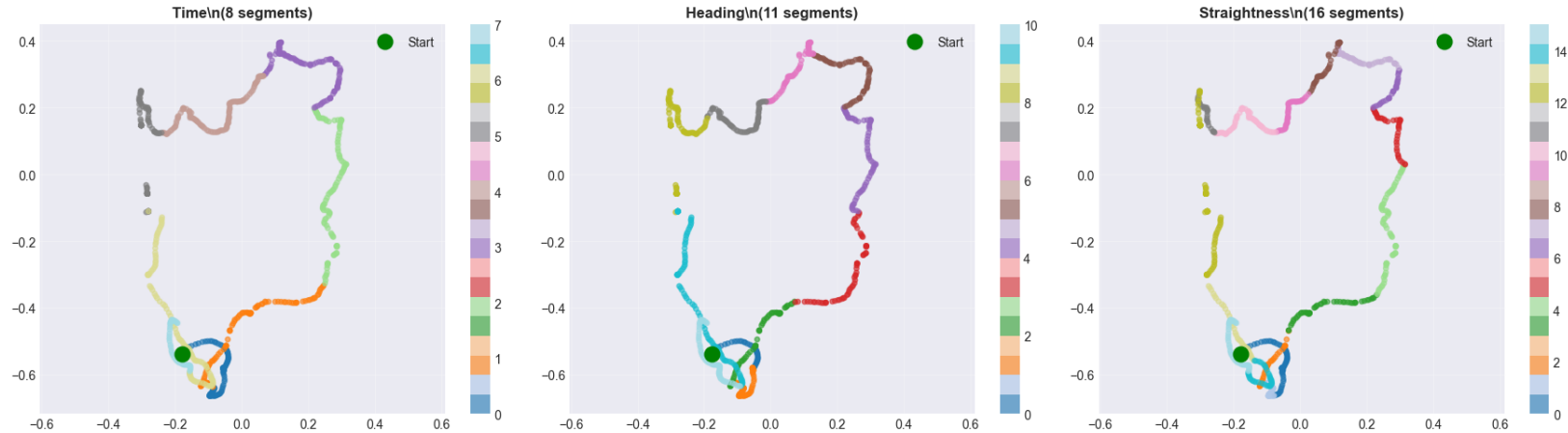
Peyrache et al (2017).

What you can try:

- Speed
- Distance to border
- Segmentation – state
- Egocentric vs allocentric: `Day5_EBR_Tutorial.ipynb`

Direction2: navigation sylabels

Day3_Part1_BasicSegmentaion



Day3_Part2_ChangePointDetection

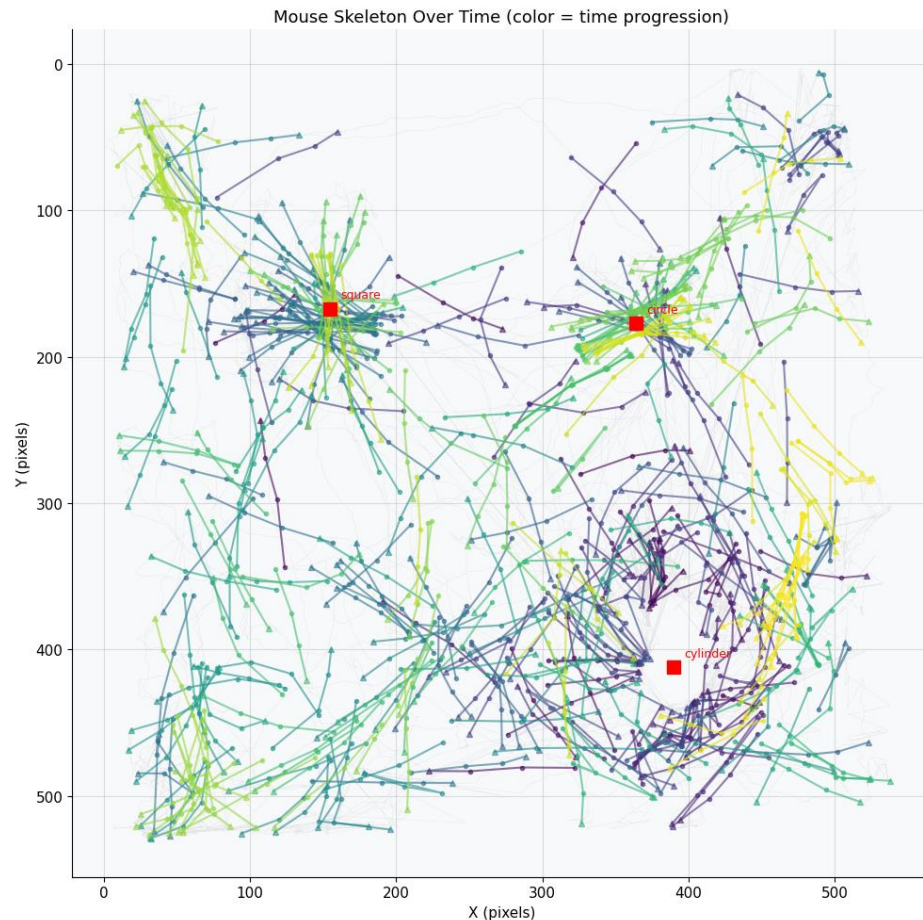
- Extract features (speed, distance to border, angular speed...)
- Clustering
- State inference by checking the feature value ...

Direction3: deeplabcut based state detection

Try to detect potential state for the novel object recognition task.

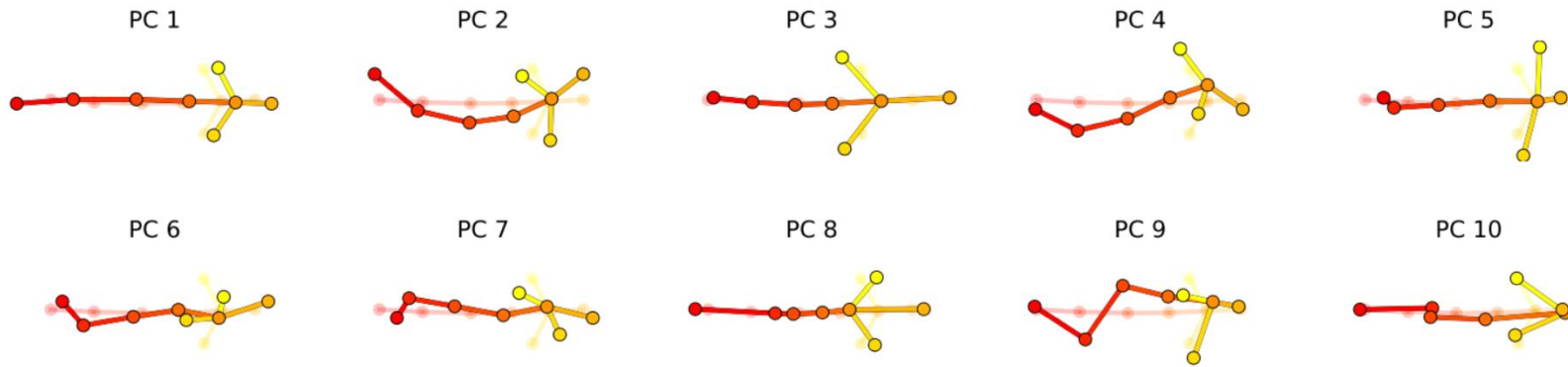
Features

- Time scale (0.5 sec?)
- PCA? - clustering – interpretation



Direction4: Keypoint moseq

Day4_keypoint_moseq_colab



Tutorial: <https://keypoint-moseq.readthedocs.io/en/latest/install.html>

Colab: https://colab.research.google.com/github/dattalab/keypoint-moseq/blob/main/docs/keypoint_moseq_colab.ipynb

Summary

Raw data

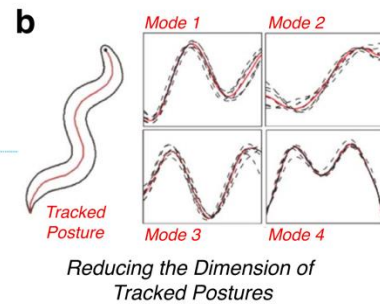
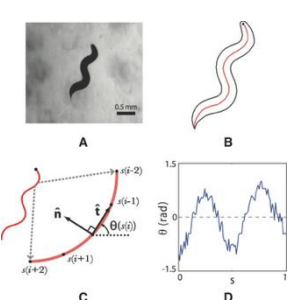
Feature

Meaningful
Timescale

Label state/syllables/
Movemes by feature

Label state by
feature

State transtion/
How state is used



Reducing the Dimension of
Tracked Postures

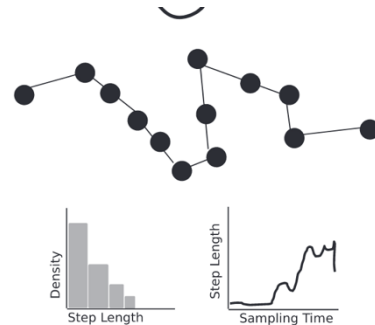
$$\theta(s) \approx \sum_{\mu=1}^K a_{\mu} u_{\mu}(s)$$

Step 1:

Actual Movement Path:



Step 2:



10 frame, 0.5sec

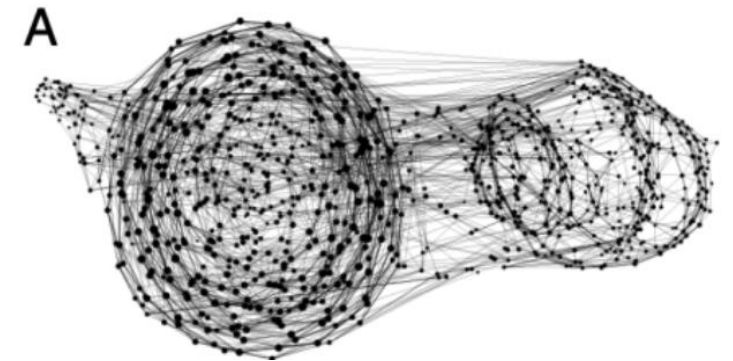
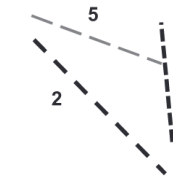
kmeans

Step 4:

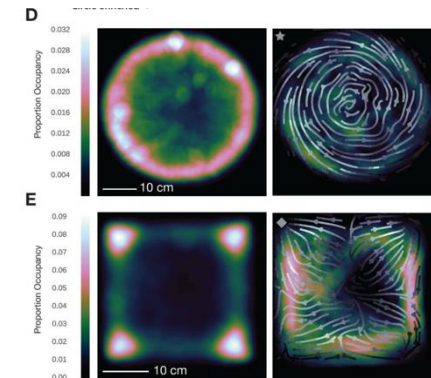
Cluster 1



Cluster 2



Outbound/inbound?



- Hierarchical structure of behavior
- Choose the suitable feature
- Meaningful scale
- Designed by your question